

Minh Vu

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Education

Gettysburg College, Bachelor of Science in Computer Science and Mathematics Dec 2026
Gettysburg, PA

- **Courses:** Data Structure and Algorithm, Database System, The Theory of Computation, Computer Vision, Ethical Hacking, Linear Algebra, Abstract Math, Abstract Algebra, Real Analysis, Chaos & Dynamical Systems
- **Achievements:** 36th place The 2024 ICPC Mid-Atlantic Regional Contest; **500+** LeetCode problems

Experience

Software Engineer Intern, VeneraAI May 2026 – Now

- Developing **FastAPI** backend services and **PostgreSQL** data models for a B2B medical AI platform supporting 50+ healthcare integration endpoints across Southeast Asia

Software Engineer Intern, CMC Global Nov 2025 – Jan 2026

- Developed 7 **Java SpringBoot** microservices for payment authorization and courier dispatch, decoupling Order and Delivery domains to support independent deployment
- Improved distributed transaction reliability by implementing **Kafka-based Saga** orchestration with automated rollback and retry mechanisms for multi-step payment workflows
- Reduced order status propagation latency from **2–3 minutes** to **under 5 seconds** by implementing an **Outbox-pattern** event pipeline and real-time **Vue.js** dashboard backed by **PostgreSQL**

Full-stack Developer, Gettysburg College Jun 2025 – Jul 2025

- Engineered a full-stack departmental event management web application using **React.js**, **Express.js**, and **AWS EC2**, automating trackings and eliminating manual spreadsheet workflows
- Designed normalized **MySQL** schemas with foreign key constraints and junction tables, eliminating duplicate registration records and improving data consistency across event workflows
- Reduced event query response times by **80%** by implementing a **Redis** layer to handle frequently accessed data

Software Engineer Intern, VinAI Jun 2024 – Aug 2024

- Developed backend services for distributed AI camera clusters, enabling real-time propagation of object recognition telemetry across 20+ simulated camera nodes with low-latency synchronization
- Engineered a **ZooKeeper-based** leader election architecture for distributed camera clusters, reducing automated failover latency to under **1.5 seconds** during simulated node outages
- Streamlined camera-to-backend communication by developing **RESTful APIs** with **Express.js** and **MongoDB** and orchestrating deployments using **Docker** and **Kubernetes**

Projects

GoScale Mail | Go, RabbitMQ, PostgreSQL, pgvector, Gemini, Redis, gRPC, Docker

- Built a self-hosted email compliance platform in **Go** that intercepted outbound emails for AI-driven PII and phishing detection, classifying threats into four severity levels while ensuring zero external data transmission
- Architected 9 microservices decoupled via **RabbitMQ** asynchronous pipelines with **OAuth2.0-secured PostgreSQL**, enabling fault-tolerant concurrent mail processing
- Engineered a **Gemini** reasoning loop over **pgvector RAG** and **Redis** behavioral graph for real-time threat scoring

Leadership

President, GBC Association for Computing Machinery January 2026 – Now

- Directed an 8-member executive board to cultivate a campus-wide computing community, orchestrating technical workshops, career development panels, and peer-led algorithmic training sessions

Technical Skills

Languages: Go, Java, Python, C++, SQL, TypeScript, JavaScript

Technologies: Spring Boot, React.js, PostgreSQL, Redis, MongoDB, PyTorch, TensorFlow

Infrastructure: AWS, Docker, Kubernetes, Kafka, RabbitMQ, ZooKeeper, gRPC